

Roll No.

TME-706

B. TECH. (ME) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
POWER PLANT ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What do you mean by load factor and plant use factor ? How is the total annual cost of electricity estimated ? (CO1)

OR

- (b) A power station of 30 MW capacity has the maximum annual demand of 25 MW. It supplies loads having maximum demands of 10 MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 45%. (CO1)

Estimate :

- (i) The average load
- (ii) The energy supplied per year
- (iii) The diversity factor
- (iv) The demand factor

P. T. O.

2. (a) Define Depreciation. What do you mean by sinking-fund method of accumulating the money for the depreciation fund ? (CO1)

OR

- (b) A plant costing ₹ 65,000 has a useful life of 15 years. Find the amount which should be saved annually to replace the equipment at the end of that time (i) by the straight-line method and (ii) by the sinking-fund method if the annual rate of compound interest is 5%. Assume that the salvage value of the equipment is ₹ 5,000. (CO1)
3. (a) Give the layout of diesel power plant. What are the advantages and disadvantages of diesel power plant ? (CO3, CO2)

OR

- (b) An impulse turbine (with a single row wheel) the mean diameter of the blades is 1.05 m and the speed is 3000 r.p.m. The nozzle angle is 18° , the ratio of blade speed to steam speed is 0.42 and the ratio of the relative velocity at outlet from the blades to that at inlet is 0.84. The outlet angle of the blade is to be made 3° less than the inlet angle. The steam flow is 10 kg/s. Draw the velocity diagram for the blades and determine the following : (CO3, CO2)
- (i) Resultant thrust on the blades
- (ii) Power developed in the blades
4. (a) Give the layout of a modern steam power plant. Explain all the circuits. (CO2, CO1)

(3)

OR

- (b) The initial investment in a power plant is ₹ 4,000 crore. After 4 years, a further ₹ 1,200 crore have to be paid and from the 5th onwards to the 12th year ₹ 400 crore per year are paid. The salvage value of the plant after 30 years will be ₹ 600 crore. The interest rate is 10%. Estimate the present worth of these payments at the time of commissioning.

(CO2, CO1)

5. (a) Explain Rankine cycle and derive the expression for its thermal efficiency. (CO2)

OR

- (b) Discuss various methods of compounding in steam turbines. (CO2)